Visual Analytics

Fall 2025 — Course Project Report

Visual Analytics of AI Capability, Compute Infrastructure, and Governance Across Countries (2010–2024)

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1 Introduction

The global expansion of artificial intelligence (AI) has led to significant differences among countries in research output, computing infrastructure, and regulatory frameworks. Understanding these differences requires integrating heterogeneous data sources and supporting exploratory, multi-dimensional analytical tasks, for which visualization analytics is well-suited.

This report introduces a visualization analytics system designed to explore AI development patterns in 30 countries between 2010 and 2024. The system integrates indicators such as AI capabilities (publications, patents), computing infrastructure (high-performance computing capabilities, energy), and governance (policy initiatives, international coordination) into a coordinated, multi-view, interactive dashboard.

The target users are policy analysts, researchers, and students and job seekers interested in global AI governance, infrastructure disparities, and finding promising jobs. Users can freely compare national situations, identify structural clusters, track time trajectories, and quantify selected statistics through direct interaction with the visualization interface

2 Related Works

This project is mainly inspired by prior work in interactive visual analytics system design, especially systems that combine dimensionality reduction, clustering, and coordinated views. A useful conceptual starting point is AI4VIS (Wu et al., 2022), which reviews how artificial intelligence and visualization can be integrated into interactive analytical systems. For this project, the most relevant idea is not simply "using AI in visualization," but designing a system where analytical methods and visual interaction work together. This is also the logic of the current dashboard: PCA, clustering, and summary statistics are not shown as isolated outputs, but are integrated into linked views that support exploration, comparison, and interpretation.

A second important influence is the literature on visual analytics systems that combine dimensionality reduction, clustering, and coordinated views. Abdullah et al. (2020) present a visual analytics system, VALENCIA, that integrates dimension reduction and cluster analysis in one environment and supports multiple subviews and interactions for exploring high-dimensional data. This is methodologically close to the design of this project. Although their application domain is healthcare, the design principle is very relevant: dimensionality reduction should not be treated as a standalone preprocessing step, but as the center of an interactive system where users can move between overview, filtering, subgroup comparison, and explanation. This is exactly how the current dashboard uses the PCA scatterplot as the main navigation view, coordinated with ranking, profile, trajectory, and summary panels.

The temporal component of the dashboard is especially related to TimeCluster (Ali et al., 2019), which was developed for the visual analysis of long time-series data. That system combines dimension reduction with connected scatterplots so that users can observe changes over time, identify patterns, and compare trajectories in a low-dimensional visual space. This directly supports the logic of the trajectory view in this project. Here, country paths from 2010 to 2024 are plotted in PCA space so that the user can see not only where a country is located, but also how it moves over time. In other words, the dashboard adopts the same general design principle: use a reduced visual space for overview, then preserve temporal continuity through trajectories.

A particularly relevant reference for the present work is Basole's Visualizing the Evolution of the AI Ecosystem (Basole, 2021). This paper is not about countries, but about the evolution of AI technology segments using venture and funding data. The connection to the present project lies in the dashboard logic of evolution. Basole visualizes AI technology segments as connected scatterplots and interprets those paths using trajectory-based measures such as path length and velocity. The key contribution is that evolution is not treated as a sequence of disconnected yearly values, but as a meaningful path in a visual space. This idea is very close to the design of the current dashboard. In this project, countries replace technology segments, and PCA space replaces Basole's ecosystem landscape, but the underlying reasoning is similar: the user should be able to compare developmental paths, detect divergence, and interpret movement patterns rather than just read static rankings. This is why the trajectory panel is not only an additional chart, but an essential design choice for understanding how national AI profiles evolve over time.

At the domain level, this project also connects to recent work on AI governance, because the dashboard does not only compare research and compute indicators, but also policy activity. In that sense, the system is designed as a visual analytics tool for understanding the relationship between capability, infrastructure, and governance, rather than a simple national ranking interface. This concern is also reflected in recent governance literature such as Batool et al. (2025), who emphasize the growing importance of governance frameworks, accountability, and policy mechanisms in AI development.

3 Dataset and Domain

3.1 Dataset Characteristics

The dataset is organized at country–year granularity and covers 30 countries over the period 2010–2024. Table 1 summarizes the scale of the final analytical dataset.

Dimension	Value
Entities	30 countries
Time range	2010–2024 (15 years)
Observations (tuples)	450 (30 × 15)
Variables (dimensions)	23 analytical variables
Angelini–Santucci Index	AS = 450 × 23 = 10,350

Table 1. Dataset scale summary.

3.2 Data Sources

The dataset was built by integrating several public sources at the country–year level. World Bank Open Data provided the main socioeconomic, research, infrastructure, and digital development indicators, including GDP per capita, R&D expenditure, researchers in R&D, internet use, broadband access, mobile subscriptions, tertiary enrollment, ICT-related exports, secure internet servers, and electricity use. The TOP500 Supercomputer List was used to derive national compute capacity through the number of systems and the summed Rmax performance in PFLOPS. AI research output was represented through the AI publications series integrated during preprocessing, while governance information was captured through yearly counts of AI policy initiatives from public policy records. Together, these sources created

a multidimensional dataset combining AI capability, compute, governance, and broader development context.

3.3 Variable Groups

To improve interpretability, the variables were organized into five thematic groups, shown in Table 2.

Group	Variables	Main source(s)
AI Capability & Innovation	AI publications; R&D % GDP; researchers per million	AI publications series; World Bank
AI Compute & Infrastructure	TOP500 systems count; TOP500 Rmax PFLOPS; power use per capita	TOP500; World Bank
AI Governance & Policy	AI policy initiatives	Public policy records
ICT Infrastructure	Internet users; fixed broadband; mobile cellular subscriptions	World Bank
Socioeconomic Context	GDP per capita; income group; OECD / GPAI membership	World Bank; membership lists

Table 2. Variable groups and corresponding indicators.

3.4 Data Preparation

Data preparation was done in Python through a staged pipeline. The workflow started with separate cleaning of the main sources, followed by harmonization at the country–year level through standardized country names, aligned year coverage, and merging into one master table. During this stage, the process had to handle inconsistent file formats, missing metadata, incomplete indicators, and variables that were initially misnamed or stored in different forms.

Missing data were treated differently according to variable type. Identifier fields and variables for which zero had a clear meaning were kept unchanged. For continuous country–year indicators that usually change gradually over time, missing values were filled within each country using linear interpolation, followed by forward-fill and backward-fill for boundary gaps. The variable ai_publications was handled separately with a KNN-based imputation method, because some countries had long missing segments that could not be filled reliably by interpolation alone. Decimal values in ai_publications were preserved because the source uses fractional counting rather than raw publication totals.

After the missing-value stage, the workflow progressed from master_data.csv to master_data_updated.csv, then to analysis_data.csv, analysis_data_filled.csv, and finally analysis_data_completed.csv. Before dimensionality reduction, all numeric variables were standardized with z-score normalization so that variables on different scales could contribute more fairly to PCA. The PCA results were exported to pca_scores.csv and then combined with KMeans labels to produce pca_scores_with_clusters_k4.csv. PC1 explain 33.4% variance and PC2 explain 20.0% variance, together capturing 53.4% of the total variance .Additional outputs such as cluster_summary_mean.csv, cluster_summary_median.csv, cluster_size.csv, and trajectory files were generated to support the final visualization.

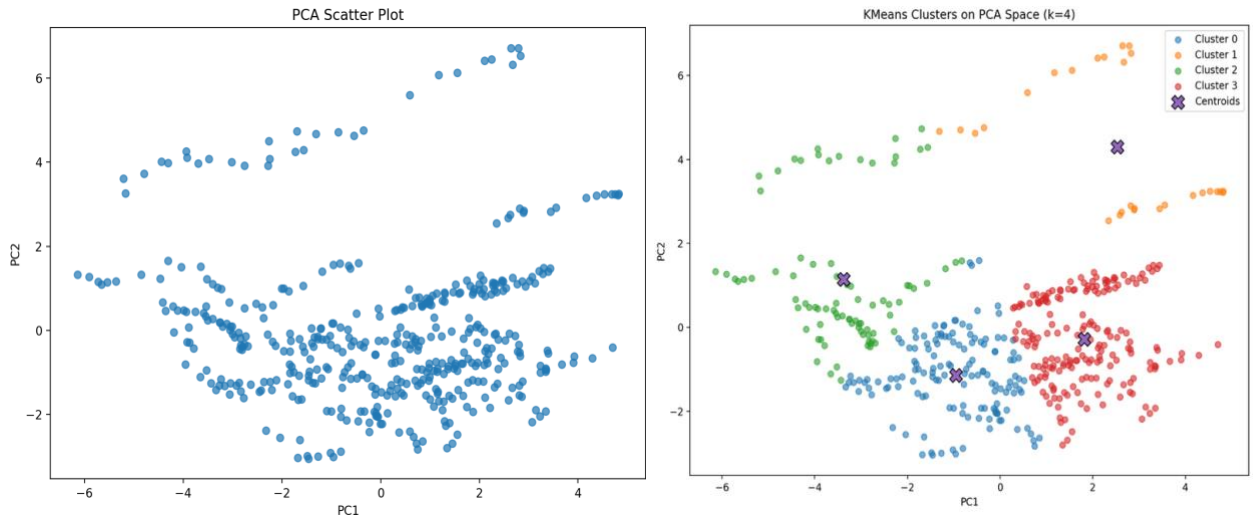


Figure 1. PCA embedding of all country–year observations and KMeans clusters on PCA space (k=4).

To help interpret the load table as table 3 illustrated PC1 can be understood as general technology and development factors, while PC2 reflects the structural contrast between AI/computing specialization and the broader maturity of the digital economy.

Rank	Variable	PC1 Loading	PC2 Loading
PC1 — Top 5 Positive Contributors			
1	Researchers per Million	**+0.3576**	-0.1668
2	GDP per Capita (USD)	**+0.3375**	-0.2655
3	Internet Users (%)	**+0.3344**	-0.2354
4	R&D Expenditure (% GDP)	**+0.3235**	-0.0203
5	Electric Power (kWh per Capita)	**+0.3150**	-0.2439
PC1 — Bottom 5 (Lowest Loadings — All Positive)			
1	Mobile Cellular per 100	+0.0239	-0.2799
2	ICT Service Exports (%)	+0.0354	+0.0370
3	Fixed Broadband per 100	+0.0928	+0.3868
4	AI Publications	+0.1058	+0.4897
5	Computer & Comm. Services (%)	+0.1812	+0.0555
PC2 — Top 5 Positive Contributors			

Rank	Variable	PC1 Loading	PC2 Loading
1	AI Publications	+0.1058	**+0.4897**
2	Fixed Broadband per 100	+0.0928	**+0.3868**
3	Top500 Systems Count	+0.2639	**+0.3616**
4	Top500 Rmax Sum (PFlops)	+0.2989	**+0.3025**
5	IT Network Security	+0.2796	**+0.1899**
PC2 — Top 5 Negative Contributors			
1	Mobile Cellular per 100	+0.0239	**-0.2799**
2	GDP per Capita (USD)	+0.3375	**-0.2655**
3	Electric Power (kWh per Capita)	+0.3150	**-0.2439**
4	Internet Users (%)	+0.3344	**-0.2354**
5	Tertiary Enrollment (Gross)	+0.2399	**-0.2062**

Note: PC1 loadings are all positive across variables, indicating a general "technology & development" factor. Negative values appear only in PC2, reflecting contrast between infrastructure/connectivity variables and AI-focused indicators.

Table 3. PCA Loadings — Top 5 Positive and Negative Contributors

PC1 mainly captures overall national strength in economic development, research capacity, digital infrastructure, and compute capability. In the same time PC2 does not represent overall strength, but rather a structural contrast between AI/compute specialization and broader digital-economic maturity.

4 Our Solution

The final system is an interactive dashboard built with HTML, CSS, JavaScript, and D3.js, while Python was used offline for data preprocessing, PCA, and KMeans. Data are loaded at runtime from CSV files, and no backend computation is required during interaction. The dashboard contains five coordinated panels: a PCA scatter plot, a ranking bar chart, a parallel coordinates view, a temporal trajectory view, and a selection summary statistics panel.

4.1 Panel 01 — PCA Scatter Plot (Main View)

The PCA scatter plot is the main navigation view of the dashboard. Each point represents a country–year observation, and the axes correspond to the first two principal components. Figure 3 shows the PCA embedding for all country–year observations, colored by geographic region, while Figure 3 shows the same space colored by KMeans cluster labels.

Color is switchable via the sidebar. Convex hulls are drawn around each cluster to aid visual grouping. The user can interact in two ways: (a) drawing a rectangular brush to select multiple countries, which immediately dims all other panels' elements and triggers a statistics recomputation; (b) clicking individual dots to add countries to the tracked set for the trajectory view. All of these selection will interacte with other 4 panels.

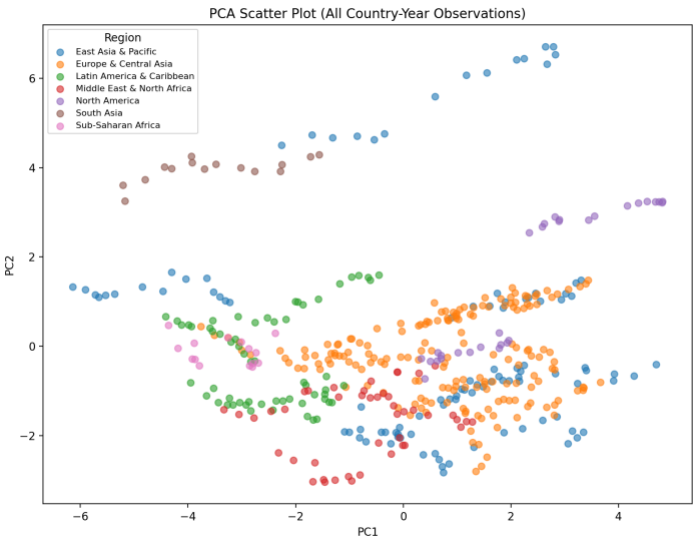


Figure 2. colored by KMeans cluster.

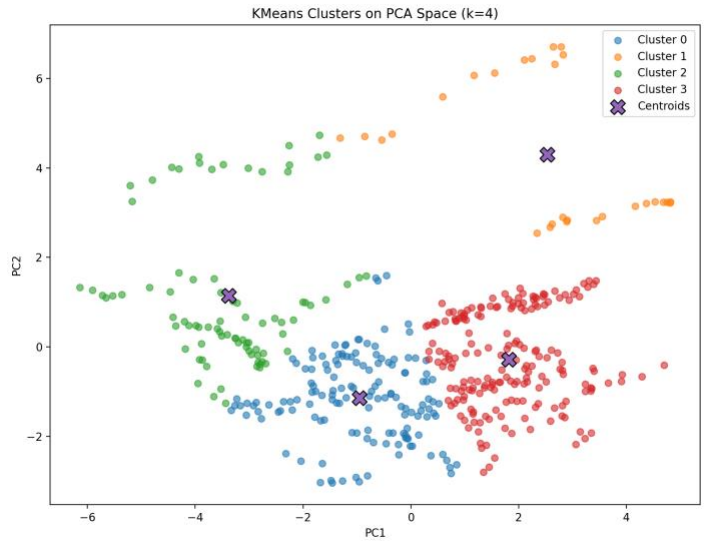


Figure 3. colored by region.

PC1 mainly captures overall national strength in development, research capacity, digital infrastructure, and compute capability. PC2 captures a structural contrast between AI/compute specialization and broader digital-economic maturity. In practice, the scatter plot is used for selection, comparison, and cluster overview; brushing a region dims the other panels and triggers a recomputation of summary statistics.

4.2 Panel 02 — Ranking Bar Chart (Coordinated View)

The ranking bar chart orders countries by a selected metric for the current year. It is coordinated with the PCA scatter plot and serves as a more explicit comparison view. Figure 4 illustrates the ranking of countries by AI publications in 2024.

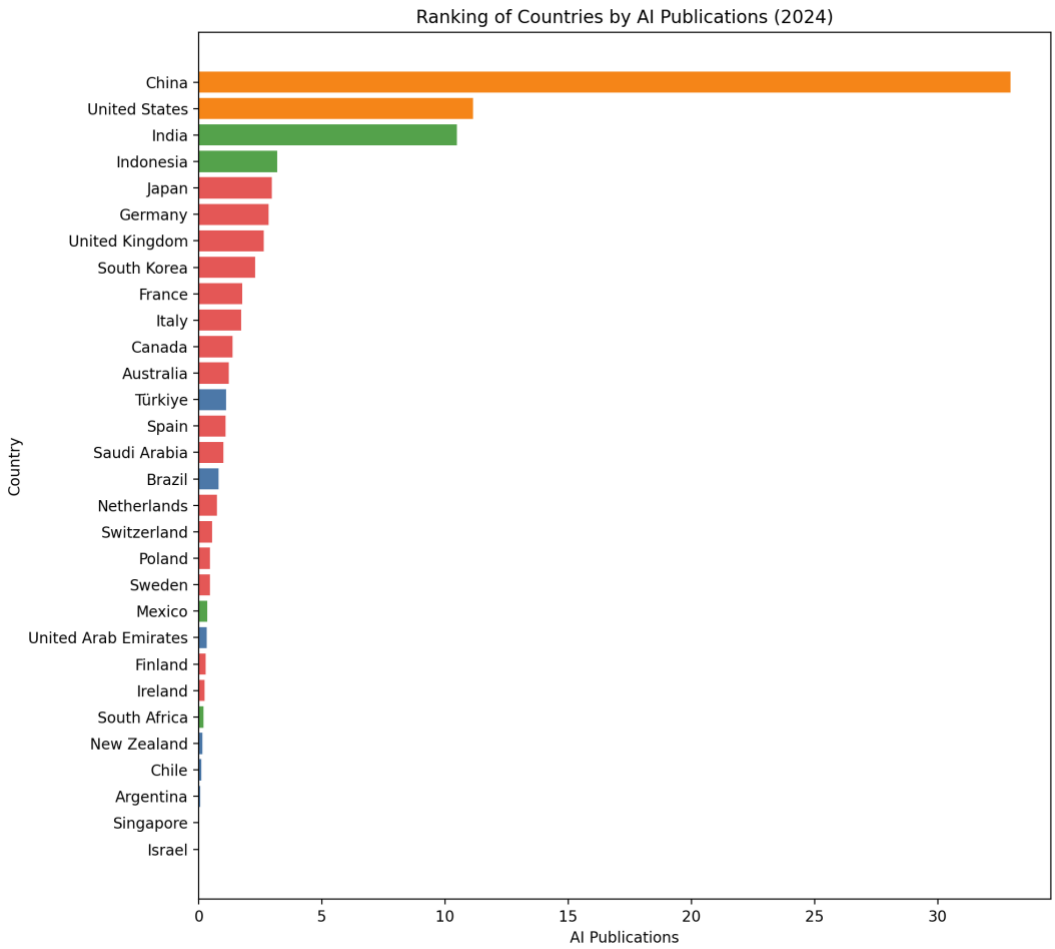


Figure 4. Ranking of countries by AI publications in 2024.

The chart helps users compare countries on specific indicators such as AI publications, AI policy initiatives, TOP500 compute, or R&D intensity. Selection is coordinated in both directions: brushing the scatter plot filters the ranking view, while clicking bars highlights the corresponding countries in the scatter plot and adds them to the trajectory tracker in panel 4 as well as panels 3 and 5 .

4.3 Panel 03 — Parallel Coordinates (Multivariate Profile View)

The parallel coordinates chart explains why countries belong to different clusters by showing normalized cluster mean profiles across key variables. Figure 5 presents the cluster-level multivariate profiles used in the dashboard.

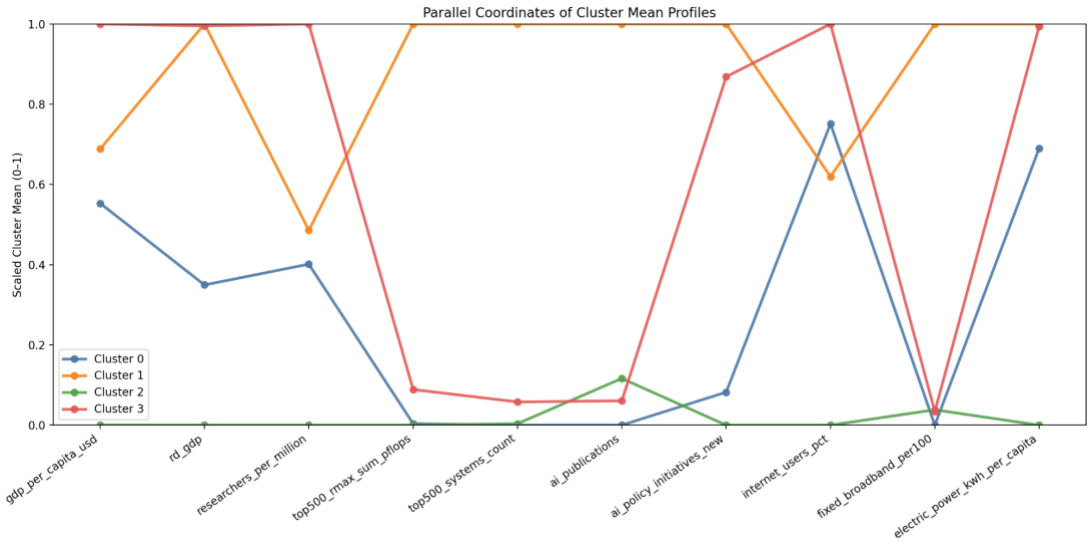


Figure 5. Parallel coordinates of cluster mean profiles.

Each line represents one cluster and summarizes variables such as GDP per capita, R&D intensity, researchers per million, internet users, TOP500 systems, TOP500 PFLOPS, AI publications, AI policy initiatives, and electricity use. This view is particularly useful for interpreting differences between clusters that may look close in the PCA scatter plot but differ in their multivariate profiles.

When a brush is active or countries are clicked, the chart highlights the cluster(s) to which the selected countries belong, dimming all others. Hovering a line reveals a tooltip showing the raw (un-normalized) cluster mean values. This view explains why countries cluster together — a user who sees that Cluster 3 leads on GDP/cap and R&D% but trails on AI Publications can infer that high-income countries are not necessarily AI research leaders. The multivariate profile view is shown in Figure 5.

4.4 Panel 04 — Temporal Trajectory View

The trajectory view shows how selected countries move in PCA space from 2010 to 2024. By default, the dashboard displays a small set of representative countries, but the user can replace them through direct interaction with the scatter plot or the ranking chart. Figure 6 shows the trajectory view for the selected countries used in the final system.

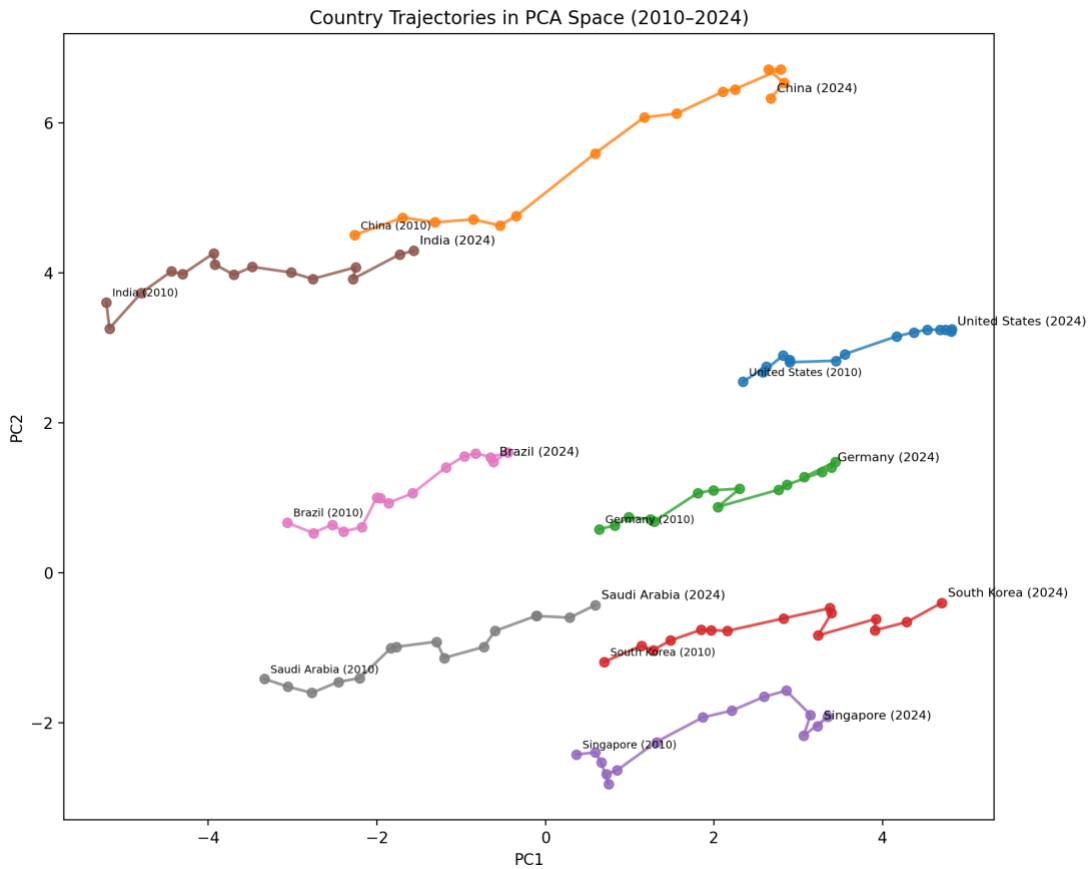


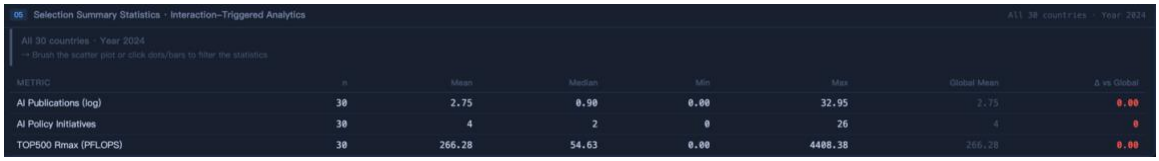
Figure 6. Country trajectories in PCA space from 2010 to 2024.

This panel supports temporal comparison and makes long-term change visible. It helps distinguish countries with rapid movement, such as China, from countries with slower or more stable paths, such as Germany or the United States.

4.5 Panel 05 — Selection Summary Statistics (Mandatory Analytics)

The selection summary statistics panel is the mandatory interaction-triggered analytics component of the project. Every time the selection state changes, the dashboard recomputes summary statistics for the selected subset relative to the current year's full dataset. The panel reports the number of selected observations, mean, median, minimum, maximum, the global mean, and the difference from the global mean for a small set of key variables.

This design makes the analytical consequence of interaction explicit. The user does not only see highlighted points or bars, but also receives immediate quantitative feedback about the selected countries. In this way, the dashboard satisfies the course requirement that at least one analytical computation must be triggered directly by visual interaction.



05 · Selection Summary Statistics · Interaction-Triggered Analytics

All 30 countries · Year 2024

All 30 countries · Year 2024
→ Brush the scatter plot or click dots/bars to filter the statistics

METRIC	n	Mean	Median	Min	Max	Global Mean	Δ vs Global
AI Publications (log)	30	2.75	0.90	0.00	32.95	2.75	0.00
AI Policy Initiatives	30	4	2	0	26	4	0
TOP500 Rmax (PFLOPS)	30	266.28	54.63	0.00	4408.38	266.28	0.00

Figure 7. Selection summary statistics panel.

5 Insights

5.1 The AI Leadership Gap

Brushing the upper-right region of the 2024 PCA scatter isolates the leading AI group, dominated by the United States and China. The summary statistics panel shows that these countries are far above the global mean in compute capacity and AI publication output. This confirms a strong leadership gap between the top group and the rest of the sample.

At the same time, China and the United States differ in how they achieve this position. China scores much higher on PC2 because it performs strongly on positive-loading variables such as AI publications, TOP500 systems, and TOP500 Rmax PFLOPS. These indicators capture scale and concentrated investment in research and infrastructure. The United States remains a global AI leader, but much of its strength is concentrated in private firms and does not always appear fully in public infrastructure metrics. In addition, some of the negative-loading variables on PC2, such as GDP per capita and broader digital maturity, are areas where the United States performs very highly, which lowers its PC2 score. This suggests that PC2 should not be read as a simple ranking of overall AI strength, but as a contrast between infrastructure-heavy, publication-heavy development and broader economic and digital maturity.

Figure 8. The AI leadership gap highlighted in the dashboard.

5.2 European Governance Leaders

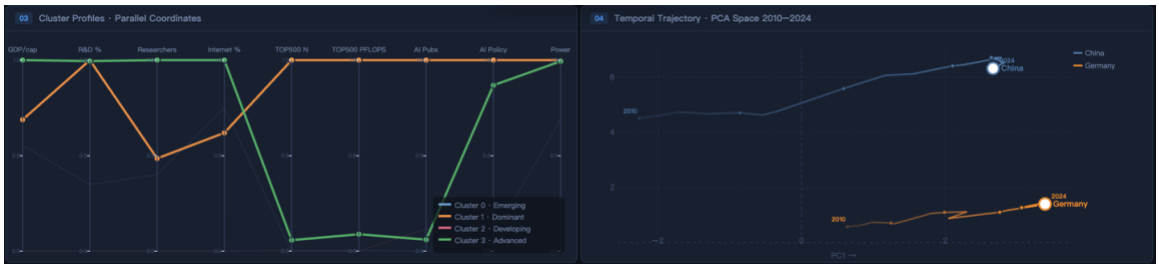
A second pattern appears when countries are ranked by AI policy initiatives. Several European countries, especially Germany, France, the United Kingdom, Finland, and Sweden, score strongly on policy activity while remaining more moderate in compute capacity. In the parallel coordinates view, this appears as a profile that peaks on the AI policy axis but stays lower on the TOP500 and publication axes. This mismatch suggests that governance leadership and compute leadership do not always coincide.

This pattern is illustrated in Figure 9.

Figure 9. European governance leaders in the interactive dashboard.

5.3 Temporal Trajectories Reveal Divergence

The trajectory view highlights different long-term development paths. China moves strongly toward the high-capability, high-compute area of the embedding, which indicates rapid growth over the period. Germany's path is shorter and more gradual, while the United States remains consistently strong with a more stable trajectory. These differences show that national AI development is path-dependent rather than uniform.



This comparison is shown in Figure 10.

Figure 10. China and Germany trajectories in PCA space (2010–2024).

5.4 Emerging Countries Show Gradual Progress

The emerging-country cluster, represented by countries such as India, Indonesia, Mexico, and South Africa, shifts gradually rightward over time in PCA space. This indicates measurable progress along the capability dimension, especially in research and digital infrastructure. Although these countries remain below the global leaders, the long-term trend is clearly positive.

6 Limitations and Constraints

This project has several limitations. First, the current dataset primarily records visible and measurable outputs, such as the number of AI publications, TOP500 system rankings, broadband access, electricity consumption, and policy initiatives. While these variables are useful, they focus more on scale, infrastructure, and public activity than on fundamental innovation. Therefore, the analysis may overestimate countries strong in large-scale infrastructure or the number of publications, while underestimating countries where AI strength is concentrated in less prominent areas.

Second, the dataset lacks several important factors. Specifically, the model does not include chip architecture patents, advanced AI hardware designs, proprietary industrial computing power beyond publicly available rankings, semiconductor supply chain control, contributions to core AI frameworks, quality of cutting-edge models, or high-impact citation performance. These variables are crucial because

they reflect original innovation, hardware efficiency, and control over the underlying technology stack. Including these variables could significantly change the relative rankings of some countries, particularly the US and China.

Finally, the results are also affected by methodological limitations. Missing values require imputation, and although the imputation strategy was carefully chosen, uncertainties remain. Principal component analysis (PCA) simplifies high-dimensional datasets to two principal axes, improving interpretability but inevitably reducing complexity. Therefore, this dashboard should be viewed as an exploratory tool rather than a final ranking of countries' AI capabilities.

7 Conclusion

This project developed a visualization and analysis system to explore the artificial intelligence capabilities, computing infrastructure, and governance status of various countries between 2010 and 2024. By integrating heterogeneous public data, applying principal component analysis (PCA) and K-means clustering algorithms, and designing a harmonized D3.js dashboard, the project meets the course's main analytical and interactive requirements.

More importantly, the system reveals interpretable patterns: significant leadership gaps exist at the top of the distribution; governance structures in European countries are relatively complex; long-term development trajectories vary across countries; and emerging countries exhibit gradual progress. The value of this dashboard lies in visualizing and facilitating the exploration of these patterns, rather than reducing them to a final ranking.

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